

# SIMATS ENGINEERING ASSESSMENT TOOL

COURSE NAME: Theory of Computation

COURSE CODE: CSA1304

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## COURSE OUTCOME COVERED

***CO1: Analyze fundamental mathematical concepts of automata theory for formal languages and decision problems.(BL:3)***

*Assessment Tool Weightage: 40%*

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**QUESTIONS: 5**  
**Duration : 1 Hour**

**Total Marks:25**

## Analytical Problem

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### **Code of Conduct:**

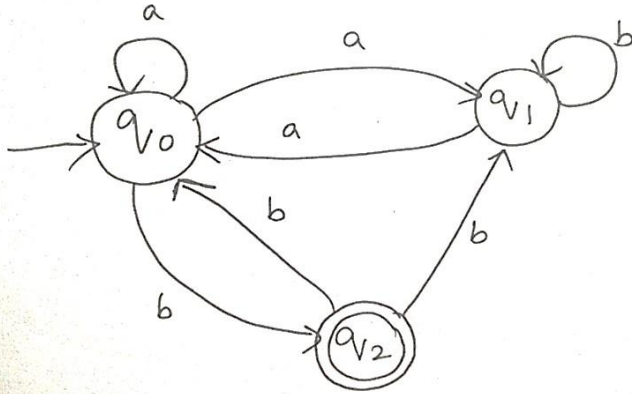
*I B.Nynika (192311075) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

*I understand that any violation of academic integrity rules will result in disciplinary action.*

**Signature of the student:**  
**B.Nynika**

# SIMATS ENGINEERING ASSESSMENT TOOL

1. Consider the NFA given below which accepts some language L over the alphabet  $\Sigma = \{a, b\}$ . Construct a DFA that represents the same language L



**SOLUTION:**

**Given NFA**

- States:  $Q = \{q_0, q_1, q_2\}$
- Alphabet:  $\Sigma = \{a, b\}$
- Start state:  $q_0$
- Final state:  $q_2$

From the diagram, the transitions are:

| State             | a              | b              |
|-------------------|----------------|----------------|
| $\rightarrow q_0$ | $\{q_0, q_1\}$ | $\{q_2\}$      |
| $q_1$             | $\{q_0\}$      | $\{q_1\}$      |
| $*q_2$            | $\emptyset$    | $\{q_0, q_1\}$ |

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### Step 1: Start State

Start subset:

$$A = \{q_0\}$$

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### Step 2: Find all reachable subsets

From  $A = \{q_0\}$

On a

$$\delta(q_0, a) = \{q_0, q_1\}$$

$\Rightarrow$

$$B = \{q_0, q_1\}$$

On b

$$\delta(q_0, b) = \{q_2\}$$

$\Rightarrow$

$$C = \{q_2\}$$

---

From  $B = \{q_0, q_1\}$

On a

$$\begin{aligned}
 &\delta(q_0, a) \cup \delta(q_1, a) \\
 &= \{q_0, q_1\} \cup \{q_0\} \\
 &= \{q_0, q_1\}
 \end{aligned}$$

So

$$B \xrightarrow{a} B$$

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On **b**

$$\begin{aligned}\delta(q_0, b) \cup \delta(q_1, b) \\ &= \{q_2\} \cup \{q_1\} \\ &= \{q_1, q_2\}\end{aligned}$$

$\Rightarrow$

$$D = \{q_1, q_2\}$$

---

**From C={q<sub>2</sub>}**

On **a**

No transition

$\Rightarrow$

$$E = \phi$$

On **b**

$$\{q_0, q_1\}$$

$\Rightarrow$

$$B$$

---

**From D={q<sub>1</sub>, q<sub>2</sub>}**

On **a**

$$\{q_0\} \cup \phi = \{q_0\}$$

$\Rightarrow$

A

On **b**

$$\{q_1\} \cup \{q_0, q_1\} = \{q_0, q_1\}$$

$\Rightarrow$

B

---

**From E=∅**

On **a**

∅

On **b**

∅

---

## Step 3: DFA Transition Table

**DFA State a b**

→A={q<sub>0</sub>}    B C

B={q<sub>0</sub>, q<sub>1</sub>}    B D

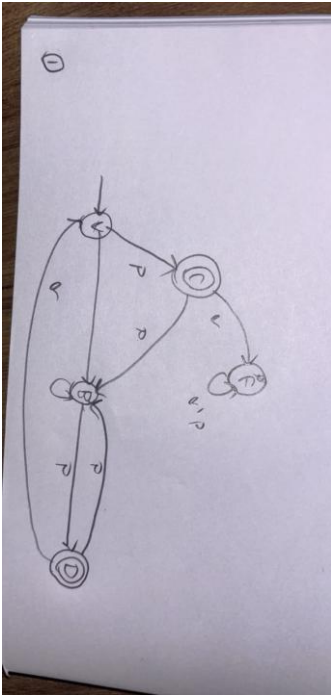
\*C={q<sub>2</sub>}    E B

\*D={q<sub>1</sub>, q<sub>2</sub>}    A B

E=∅    E E

(Accepting states are those containing  $q_2$ : C and D.)

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2. Let  $L$  be the language of words of length at least 2 whose last two letters are different from each other. For example  $abaab$  is in  $L$  but not  $bb$ , Construct an NFA for  $L$ , Then draw the equivalent DFA

SOLUTION:

### Step 1: Idea

The last two symbols should be either:

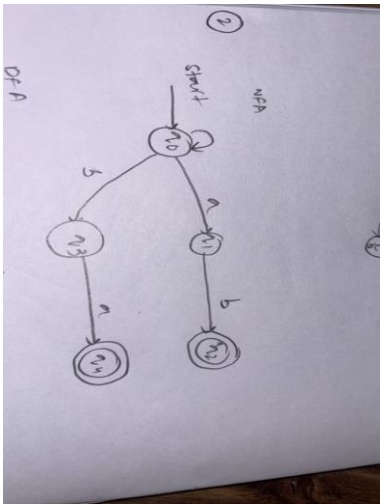
- **ab**
- **ba**

So the NFA accepts strings ending with **ab** or **ba**.

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### Step 2: NFA Diagram

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Where

- $q_0$  = Start state
- $q_2$  = Final state (ends with ab)
- $q_4$  = Final state (ends with ba)

---

## Step 3: Transition Table

| State             | a              | b              |
|-------------------|----------------|----------------|
| $\rightarrow q_0$ | $\{q_0, q_1\}$ | $\{q_0, q_3\}$ |
| $q_1$             | $\emptyset$    | $\{q_2\}$      |
| $*q_2$            | $\emptyset$    | $\emptyset$    |
| $q_3$             | $\{q_4\}$      | $\emptyset$    |
| $*q_4$            | $\emptyset$    | $\emptyset$    |

---

## Step 4: Formal Definition

$$M = (Q, \Sigma, \delta, q_0, F)$$

Where

- $Q = \{q_0, q_1, q_2, q_3, q_4\}$
- $\Sigma = \{a, b\}$
- Start state

$$q_0$$

Final states

$$F = \{q_2, q_4\}$$

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## Step 5: Equivalent DFA

The DFA only needs to remember the **last symbol**.

**DFA States**

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## State Meaning

- q0 Start
- q1 Last symbol = a
- q2 Last symbol = b
- \*q3 Ends with ab
- \*q4 Ends with ba

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## DFA Transition Table

State a b

→q0 q1 q2

q1 q1 q3

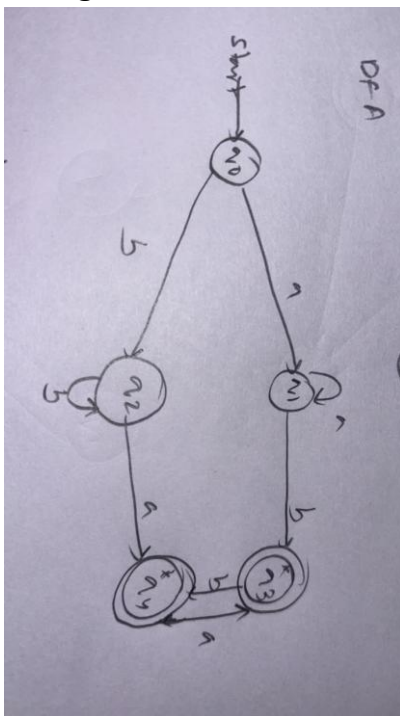
q2 q4 q2

\*q3 q4 q2

\*q4 q1 q3

---

## DFA Diagram



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## Final Answer (Exam)

### NFA Transition Table

State a b

→q0 {q0,q1} {q0,q3}

q1 ∅ {q2}

\*q2 ∅ ∅

q3 {q4} ∅

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State a      b

\*q4    $\emptyset$        $\emptyset$

### DFA Transition Table

State a   b

$\rightarrow q_0$     $q_1$     $q_2$

$q_1$        $q_1$     $q_3$

$q_2$        $q_4$     $q_2$

\* $q_3$        $q_4$     $q_2$

\* $q_4$        $q_1$     $q_3$

3. Design a Nondeterministic Finite Automata (NFA) that takes a binary string as input and accepts the string only if the last two digits are the same (either 00 or 11). Write the formal definition of the NFA. Also show that the string **01011** is accepted by the NFA

SOLUTION:

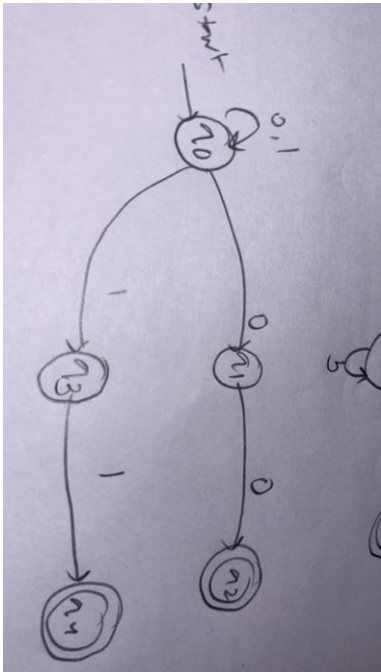
#### Step 1: Idea

A string is accepted if its **last two symbols** are:

- 00
- 11

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#### Step 2: NFA Diagram



Loop at  $q_0$  on both 0 and 1.

#### Loops at $q_0$ :

- $q_0 \xrightarrow{0} q_0$
- $q_0 \xrightarrow{1} q_0$

These loops allow the NFA to read any prefix before checking the last two symbols.

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## Step 3: Transition Table

| State             | 0              | 1              |
|-------------------|----------------|----------------|
| $\rightarrow q_0$ | $\{q_0, q_1\}$ | $\{q_0, q_3\}$ |
| $q_1$             | $\{q_2\}$      | $\emptyset$    |
| $*q_2$            | $\emptyset$    | $\emptyset$    |
| $q_3$             | $\emptyset$    | $\{q_4\}$      |
| $*q_4$            | $\emptyset$    | $\emptyset$    |

---

## Step 4: Formal Definition

$$M = (Q, \Sigma, \delta, q_0, F)$$

where

**States**

$$Q = \{q_0, q_1, q_2, q_3, q_4\}$$

**Alphabet**

$$\Sigma = \{0, 1\}$$

**Transition Function**

Given in the table above.

**Start State**

$$q_0$$

**Final States**

$$F = \{q_2, q_4\}$$

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## Step 5: Check whether 01011 is accepted

Input:

01011

The last two symbols are

11

The NFA can follow this path:

$q_0 \xrightarrow{0} q_0$

$q_0 \xrightarrow{1} q_0$

$q_0 \xrightarrow{0} q_0$

$q_0 \xrightarrow{1} q_3$

$q_3 \xrightarrow{1} q_4$

Since  $q_4$  is a final state,

|                   |
|-------------------|
| 01011 is Accepted |
|-------------------|



# SIMATS ENGINEERING ASSESSMENT TOOL

## Final Answer (Exam)

### Formal Definition

$$M = (Q, \Sigma, \delta, q_0, F)$$

where

- $Q = \{q_0, q_1, q_2, q_3, q_4\}$
- $\Sigma = \{0, 1\}$
- Start state =  $q_0$
- Final states =  $\{q_2, q_4\}$

### Transition Table

| State             | 0              | 1              |
|-------------------|----------------|----------------|
| $\rightarrow q_0$ | $\{q_0, q_1\}$ | $\{q_0, q_3\}$ |
| $q_1$             | $\{q_2\}$      | $\emptyset$    |
| $*q_2$            | $\emptyset$    | $\emptyset$    |
| $q_3$             | $\emptyset$    | $\{q_4\}$      |
| $*q_4$            | $\emptyset$    | $\emptyset$    |

### Result

|                                            |
|--------------------------------------------|
| 01011 is accepted because it ends with 11. |
|--------------------------------------------|

4. Consider the NFA given below which accepts the language representing the set of all strings having 001 as a substring over the alphabet  $\Sigma = \{0, 1\}$ . Construct a DFA that represents the same language.

SOLUTION:

#### Step 1: Idea

The DFA should remember how much of the substring **001** has been matched.

States:

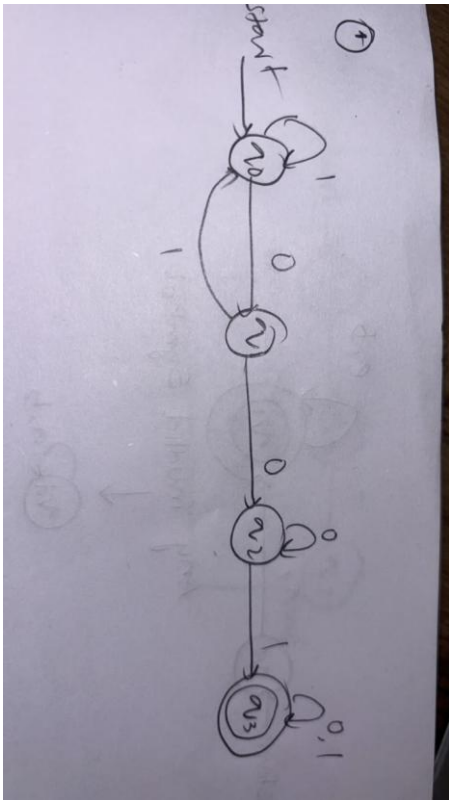
- $q_0$  – No part of "001" matched.
- $q_1$  – Matched **0**.
- $q_2$  – Matched **00**.
- $q_3$  – Matched **001** (Final State).

Once **001** is found, the DFA remains in the final state for all further inputs.

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#### Step 2: State Diagram

# SIMATS ENGINEERING ASSESSMENT TOOL



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## Step 3: Transition Table

Present State 0 1

|                   |             |
|-------------------|-------------|
| $\rightarrow q_0$ | $q_1 \ q_0$ |
| $q_1$             | $q_2 \ q_0$ |
| $q_2$             | $q_2 \ q_3$ |
| $*q_3$            | $q_3 \ q_3$ |

---

## Step 4: Formal Definition

$$M = (Q, \Sigma, \delta, q_0, F)$$

where

**States**

$$Q = \{q_0, q_1, q_2, q_3\}$$

**Alphabet**

$$\Sigma = \{0, 1\}$$

**Start State**

$$q_0$$

**Final State**

$$F = \{q_3\}$$

---

## SIMATS ENGINEERING ASSESSMENT TOOL

### Step 5: Example

Check whether

100101

is accepted.

#### Input Current State

Start  $q_0$

1  $q_0$

0  $q_1$

0  $q_2$

1  $q_3$  ✓

0  $q_3$

1  $q_3$

Since the machine reaches  $q_3$ , the string is **Accepted**.

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### Final Answer (Exam)

#### Transition Table

##### State 0 1

$\rightarrow q_0$   $q_1$   $q_0$

$q_1$   $q_2$   $q_0$

$q_2$   $q_2$   $q_3$

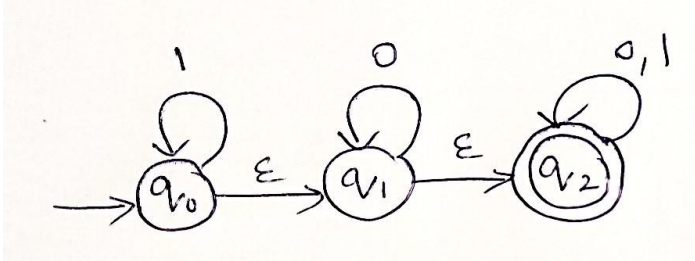
\* $q_3$   $q_3$   $q_3$

**Final State:**  $q_3$

---

The above DFA accepts **all binary strings containing the substring 001**.

5. A NFA with  $\epsilon$ -transitions is given below. Find  $\epsilon$ -closure for each state and Design an equivalent DFA which accepts the same language. Draw its transition table and mark the initial and final states



SOLUTION:

**Step 1: Given  $\epsilon$ -NFA**

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States:

$$Q = \{q_0, q_1, q_2, q_3\}$$

Alphabet:

$$\Sigma = \{a, b\}$$

Start state:

$$q_0$$

Final state:

$$q_3$$

---

## Step 2: Find $\epsilon$ -Closures

The  $\epsilon$ -closure of a state is the set of all states reachable using only  $\epsilon$ -transitions.

### State $\epsilon$ -Closure

$$q_0 \quad \{q_0, q_1, q_2\}$$

$$q_1 \quad \{q_1\}$$

$$q_2 \quad \{q_2\}$$

$$q_3 \quad \{q_3\}$$

---

## Step 3: Initial DFA State

The start state is the  $\epsilon$ -closure of  $q_0$ .

$$A = \{q_0, q_1, q_2\}$$

---

## Step 4: Subset Construction

From **A** =  $\{q_0, q_1, q_2\}$

On **a**

$$\delta(A, a) = \{q_1, q_3\}$$

Take  $\epsilon$ -closure:

$$B = \{q_1, q_3\}$$

On **b**

$$\delta(A, b) = \{q_2\}$$

Take  $\epsilon$ -closure:

$$C = \{q_2\}$$

---

From **B** =  $\{q_1, q_3\}$

On **a**

$$\{q_3\}$$

$$\rightarrow D = \{q_3\}$$

On **b**

No transition

$$\rightarrow E = \emptyset$$

---

From **C** =  $\{q_2\}$

On **a**

No transition

$$\rightarrow E$$

On **b**

$$\{q_3\}$$

$$\rightarrow D$$

---

From **D** =  $\{q_3\}$

On **a**

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$\emptyset$   
On **b**  
 $\emptyset$   
 $\rightarrow E$

From  $E = \emptyset$

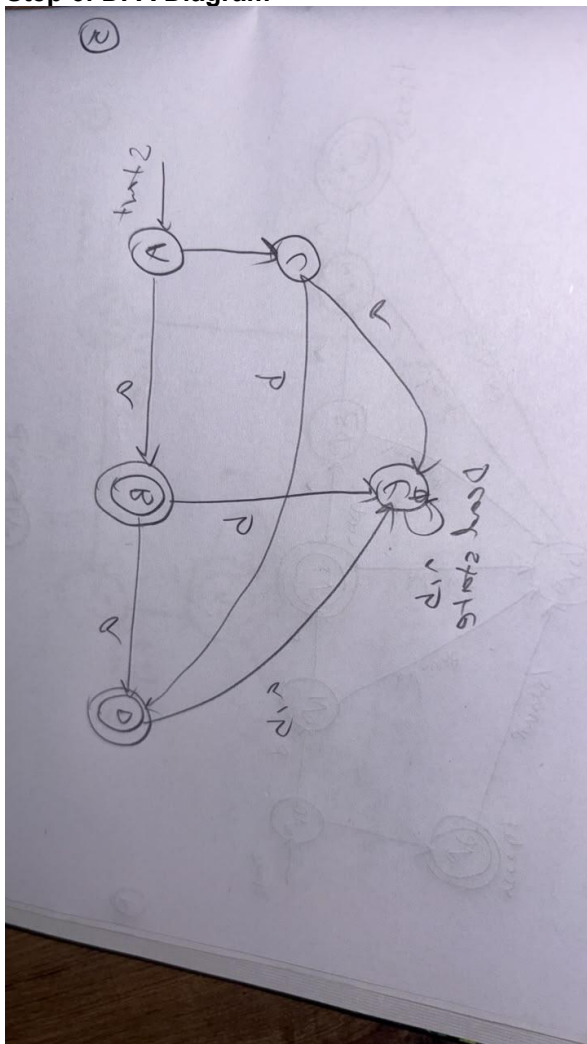
On **a**  
 $E$   
On **b**  
 $E$

## Step 5: DFA Transition Table

| DFA State                           | a | b |
|-------------------------------------|---|---|
| $\rightarrow A = \{q_0, q_1, q_2\}$ | B | C |
| $*B = \{q_1, q_3\}$                 | D | E |
| $C = \{q_2\}$                       | E | D |
| $*D = \{q_3\}$                      | E | E |
| $E = \emptyset$                     | E | E |

**Accepting states: B and D** (because they contain  $q_3$ ).

## Step 6: DFA Diagram



Where:

- **A** =  $\{q_0, q_1, q_2\}$  (Start state)
- **B** =  $\{q_1, q_3\}$  (Final)
- **C** =  $\{q_2\}$
- **D** =  $\{q_3\}$  (Final)
- **E** =  $\emptyset$  (Dead state)

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Final Answer (Exam)

$\epsilon$ -Closure Table

State  $\epsilon$ -Closure

|       |                     |
|-------|---------------------|
| $q_0$ | $\{q_0, q_1, q_2\}$ |
| $q_1$ | $\{q_1\}$           |
| $q_2$ | $\{q_2\}$           |
| $q_3$ | $\{q_3\}$           |

DFA Transition Table

State a b

|                 |   |   |
|-----------------|---|---|
| $\rightarrow A$ | B | C |
| *B              | D | E |
| C               | E | D |
| *D              | E | E |
| E               | E | E |

Final States

$B, D$

# **SIMATS ENGINEERING ASSESSMENT TOOL**